

FIELDNODE · REQUIREMENTS

FieldNode requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/03-requirements.md @ c7559da		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status against each
0.1	2026-09-25	Amish Chadha	Initial scaffold

FieldNode requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish, and will be checked by calculation at TRL 3. The status column gives the concept's standing from the first-order estimates in FND-PRC-001; "Met (estimate)" means met on paper only.

Table 1. Requirements and concept status

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Weather protection	Enclosure IP65 or better with membrane vent; all penetrations sealed glands or rated connectors	Enclosure rating; later spray test	Met by design (rated stock enclosure); gland and connector sealing unverified
R2	Operating temperature	-20 to +45 °C ambient; electronics rated -20 to +70 °C inside the enclosure	Datasheets; thermal estimate	At risk: interior temperature in full sun unverified (see R3)
R3	Interior temperature in sun	60 °C or less inside the enclosure at 45 °C ambient, full sun, still air	Thermal calculation	Not shown. Panel shading helps, but the estimate has no margin to claim
R4	Safe charging window	Cell charging blocked below 0 °C and above 45 °C by an NTC on the cell	Charger design review	Met by design
R5	Energy neutral in the worst month	Daily harvest exceeds full-load demand at 2 peak sun hours	Energy budget	Met (estimate): about 7.8 Wh stored against 3.1 Wh drawn
R6	Autonomy without sun	5 days or more at full sensor allowance, starting from 80 % charge usable	Energy budget	Met (estimate): about 5.0 days, no margin
R7	Sensor power allowance	100 mW average or more available to the sensor ports	Energy budget	Met (estimate): about 115 mW
R8	Radio link	LoRaWAN uplink to a gateway 2 km away in suburban terrain at SF9 or faster	Link budget; later field test	Unverified (range figures are estimates)
R9	Airtime within fair use	30 s uplink airtime per day or less at the default interval	Airtime calculation	Met at SF7 to SF9 with a 20-byte payload every 15 min (about 24 s at SF9). Not met at SF10 or slower unless the interval is 30 min or longer
R10	Store and forward	30 days or more of readings kept on the node when the link is down	Memory calculation	Met by design (SPI flash); sizing to confirm
R11	Standard sensor interface	Two sealed ports carrying I2C, UART or RS-485, one analog input and switched 3.3 V, 5 V and 12 V rails	Pinout review with adopting projects	Proposed; pinout awaiting Amish and project teams
R12	Mounting and install	Fits 40 to 60 mm poles and flat walls; one person installs in 15 min or less with hand tools	Design review; later timed install	Met by design (band clamps and back plate); time unverified
R13	Wind	Survives 35 m/s (126 km/h) gusts on the panel without loosening	Load calculation	Unverified: about 50 N on the panel (estimate)
R14	Mass	2.5 kg or less including panel and mounts	Massing model, then weighing	Met (estimate): about 1.7 kg

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R15	Serviceable	Cell replaced in 10 min or less with a screwdriver; no soldering in the field	Design review	Met by design
R16	Cost	Parts \$150 or less per node at quantity 1, gateway excluded	Priced BOM	Met (indicative): about \$126
R17	Open and independent	All design files under CERN-OHL-S-2.0 and MIT; works with any LoRaWAN network server, no closed cloud	Design review	Met by design
R18	Privacy pass-through	The core sends only what the sensor firmware passes to it; no images or audio leave the node	Firmware design review	Met by design; each project states its own rule

■ Requirements not met or at risk

- **R3 (interior temperature) not shown.** A light-colored box shaded by the panel may still exceed 60 °C in still air at 45 °C ambient. This drives cell life and the charge window.
- **R9 (airtime) not met at SF10 or slower.** At SF10 a 20-byte uplink takes about 0.45 s, so 96 uplinks a day use about 43 s. Distant nodes must report every 30 min or less often.
- **R6 (autonomy)** is met with no margin; a heavier sensor load or an older cell misses it.
- **R8 (range)** and **R13 (wind)** are unverified estimates.

■ Assumptions

- Worst month 2 peak sun hours on the tilted panel; panel, dust and angle losses 20 %; MPPT charger 85 % efficient at 1S LiFePO4 voltages; cell charge efficiency 95 %; rail converters 90 %. All estimates.
- LiFePO4 cell 3.2 V, 6 Ah (about 19 Wh), 80 % usable.
- Uplink payload 20 bytes plus 13 bytes of LoRaWAN overhead, 125 kHz bandwidth, coding rate 4/5, 8-symbol preamble.
- Core consumption (controller asleep, radio bursts, power board quiescent) under 0.01 Wh/day; estimate from SX1262 datasheet figures ([Semtech SX1261/2 datasheet](#)).